Question ID: 42c71eb5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

$(2x+5)^2-(x-2)+2(x+3)$

Which of the following is equivalent to the expression above?

Answer

A. $4x^2+21x+33$

B. $4x^2+21x+29$

C. $4x^2+x+29$

D. $4x^2+x+33$

Correct Answer: A

Rationale

Choice A is correct. The given expression can be rewritten as $(2x+5)^2+(-1)(x-2)+2(x+3)$. Applying the distributive property, the expression $(-1)(x-2)+2(x+3)$can be rewritten as $-1(x)+(-1)(-2)+2(x)+2(3)$, or $-x+2+2x+6$. Adding like terms yields $x+8$. Substituting $x+8$ for $(-1)(x-2)+2(x+3)$ in the given expression yields $(2x+5)^2+x+8$. By the rules of exponents, the expression $(2x+5)^2$ is equivalent to $(2x+5)(2x+5)$. Applying the distributive property, this expression can be rewritten as $2x(2x)+2x(5)+5(2x)+5(5)$, or $4x^2+10x+10x+25$. Adding like terms gives $4x^2+20x+25$. Substituting $4x^2+20x+25$ for $(2x+5)^2$ in the rewritten expression yields $4x^2+20x+25+x+8$, and adding like terms yields $4x^2+21x+33$.

Choices B, C, and D are incorrect. Choices C and D may result from rewriting the expression $(2x+5)^2$ as $4x^2+25$, instead of as $4x^2+20x+25$. Choices B and C may result from rewriting the expression $-(x-2)$ as $-x-2$, instead of $-x+2$.